Project 26: Card-Fraud Triage: GMM Density Scoring vs Autoencoder Anomaly Detector

A Major Project Report
Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Technology
in
Computer Science & AI

by

Yashkumar Nimje	**2401020113**
Saksham Narotra	**2401010402**
Rudra Pratap Singh Choudhary	**2401010394**
Abhigya Sachdeva	**2401020003**
Somraj Nandi	**2401020070**

(Group ID: Project 26) Supervisor:

Shobhit Johri

Department of Computer Science & AI
Rishihood University
Sonipat_131021 (India)

27 October 2026

Certificate

Department of Computer Science & AI
Rishihood University, Sonipat

It is certified that the work contained in the project report entitled "Project 26: Card-Fraud Triage: GMM Density Scoring vs Autoencoder Anomaly Detector" by the following students has been carried out under my/our supervision and that this work has not been submitted elsewhere for a degree.

Yashkumar Nimje	2401020113
Saksham Narotra	2401010402
Rudra Pratap Singh Choudhary	2401010394
Abhigya Sachdeva	2401020003
Somraj Nandi	2401020070

Date: Shobhit Johri

This project report entitled "Project 26: Card-Fraud Triage: GMM Density Scoring vs Autoencoder Anomaly Detector" submitted by the group Project 26 is approved for the degree of Bachelor of Technology.
The viva-voce examination has been held on _____.

Supervisor(s) Examiner(s)

Declaration

Rishihood University
27 October 2026

I/We declare that this written submission represents my/our ideas in my/our own words and where others' ideas or words have been included, I/We have adequately cited and referenced the original sources. I/We declare that I/We have properly and accurately acknowledged all sources used in the production of this report. I/We also declare that I/We have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I/We understand that any violation of the above will be a cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Yashkumar Nimje	2401020113	_____
Saksham Narotra	2401010402	_____
Rudra Pratap Singh Choudhary	2401010394	_____
Abhigya Sachdeva	2401020003	_____
Somraj Nandi	2401020070	_____

Acknowledgements

Rishihood University
27 October 2026

We express our sincere gratitude to our instructors, **Shobhit Sir** , for their constant guidance, constructive critique, and technical leadership throughout the execution of **Project 26: Card Fraud Triage (Phase 1)**. Their rigorous feedback on probabilistic modeling, data leakage prevention, and imbalanced evaluation design significantly elevated the quality of this work.

We gratefully acknowledge the Machine Learning Group (MLG) at Université Libre de Bruxelles (ULB) and the Kaggle platform for curating and making publicly accessible the benchmark Credit Card Fraud Detection dataset that formed the experimental basis of our investigation.

This project was developed through the close collaboration of all five team members:

- **Yashkumar Nimje** (2401020113) — Team Lead & Data Architecture
- **Saksham Narotra** (2401010402) — Feature Engineering & Preprocessing
- **Rudra Pratap Singh Choudhary** (2401010394) — Probabilistic Modeling (GMM Core)
- **Abhigya Sachdeva** (2401020003) — Model Selection & Robustness
- **Somraj Nandi** (2401020070) — Evaluation Framework & Decision Systems (Report Assembly)

Finally, we extend our heartfelt thanks to our families, peers, and institution for their unwavering encouragement throughout this semester.

Yashkumar Nimje	2401020113
Saksham Narotra	2401010402
Rudra Pratap Singh Choudhary	2401010394
Abhigya Sachdeva	2401020003
Somraj Nandi	2401020070

Abstract

Financial fraud imposes severe monetary and reputational damages on card payment networks globally. In benchmark transaction distributions, fraudulent events constitute an extreme rarity (less than 0.2% base rate), creating a severe class imbalance that renders standard supervised classification metrics and models ineffective. This report details the design, implementation, and empirical evaluation of **Project 26: Card Fraud Triage (Phase 1)**, an unsupervised probabilistic anomaly detection system applied to the benchmark European Cardholder dataset comprising 284,807 transactions.

To completely prevent lookahead bias and data leakage, transactions are strictly partitioned chronologically by timestamp into Training (earliest 70%, 198,608 transactions), Validation (subsequent 15%, 42,558 transactions), and Held-Out Test (final 15%, 42,560 transactions) subsets. The machine learning pipeline features robust logarithmic transformation $\log(1 + x)$ on the transaction amount and retains the 28 orthogonal principal components (V_1–V_{28}). A Gaussian Mixture Model (GMM) with full covariance matrices and regularization (`reg_covar` = 10^{-6}) is trained exclusively on legitimate transactions (Class = 0), learning the nominal probability density function $p(\mathbf{x} \mid \hat{\boldsymbol{\theta}})$ via the Expectation-Maximization (EM) algorithm. Model complexity selection is guided by Bayesian Information Criterion (BIC), identifying $K = 15$ as the optimal component count. Unsupervised anomaly scores are derived using the negative log-likelihood $-\ln p(\mathbf{x} \mid \hat{\boldsymbol{\theta}})$.

The evaluation framework, authored by Member 5 (Somraj Nandi), focuses on operational triage constraints and threshold calibration strictly on validation data. On the holdout test set, the production GMM achieves an Average Precision (PR-AUC) of **0.6432** and an ROC-AUC of **0.9512**, vastly outperforming the random-guessing baseline (PR-AUC 0.0012). Under operational review queue constraints, the Top-100 queue catches **59.62%** of all test fraud with a precision of **31.00%** (a 253.7× enrichment over the base rate), while the Top-500 queue detects **80.77%** of fraud with an 8.40% precision. A three-tier triage system is established with automated approval for low-risk transactions (score $<$ 33.37), analyst manual review queues (33.37 $\leq$ score $<$ 116.88), and high-

confidence automated blocking (score $\geq$ 116.88). Finally, theoretical limitations of high-dimensional Gaussian modeling are analyzed, establishing the rationale for Phase 2 deep autoencoders and hybrid density scoring.

Table of Contents

Acknowledgements **iii**

Abstract **v**

List of Figures **ix**

List of Tables **x**

1 Introduction **1**

1.1 Problem Motivation and the Fraud Triage Paradigm 1

1.2 Dataset Specification and Hygiene 2

1.3 Chronological Splitting Protocol 2

1.4 Team Roles and Technical Contributions 3

1.5 Report Organization 4

2 Literature Review **6**

2.1 Density-Based Anomaly Detection in Financial Fraud 6

2.2 Gaussian Mixture Models and the EM Algorithm 6

2.3 Information Criteria for Model Selection: AIC vs. BIC 7

2.4 Evaluation Metrics for Severe Class Imbalance 8

2.5 Operational Review Capacity and Triage Thresholding 9

2.6 Deep Learning and Hybrid Architectures (Bridge to Phase 2) 9

3 Problem Formulation and Proposed Approach **10**

3.1 System Architecture and Data Pipeline 10

3.2 Data Preprocessing and Feature Engineering (M1 & M2) 11

3.2.1 Chronological Partitioning Summary (M1) 11

3.2.2 Logarithmic Transformation of Transaction Amount (M2) 12

3.2.3 Treatment of Principal Components V_1–V_{28} and Scaler Selection (M2) 12

3.3 Gaussian Mixture Modeling and Anomaly Scoring (M3) 13
3.3.1 Mathematical Formulation 13
3.3.2 EM Algorithm and Covariance Regularization 14
3.3.3 Log-Likelihood Anomaly Scoring Engine 14
3.4 Model Selection and Experimental Ablations (M4) 15
3.4.1 Hyperparameter Optimization via AIC and BIC Sweeps 15
3.4.2 Covariance Matrix Topology Ablation 16
3.4.3 Dimensionality Reduction Ablation 17

4 Evaluation Methodology and Results 18
4.1 Evaluation Methodology for Extreme Imbalance 18
4.1.1 Precision-Recall AUC and Average Precision (AP) 18
4.1.2 ROC-AUC 19
4.1.3 Recall at Fixed False Positive Rate (FPR Budgets) 19
4.1.4 Precision at Top-K (Operational Review Capacities) 19
4.2 Validation-Calibrated Threshold Selection 20
4.3 Final Holdout Test-Set Evaluation Results 20
4.3.1 Continuous Ranking Performance (PR and ROC Curves) 21
4.3.2 Test Set Operational Threshold Verification 22
4.4 Operational Review Capacity and Triage Analysis 22
4.4.1 Analyst Review Capacity Queue Benchmark (Top-K) 22
4.4.2 Multi-Tier Triage Routing Architecture 23
4.4.3 Business Impact: False Alarms vs. Missed Fraud Trade-off 23

5 Limitations, Conclusion, and Future Work 25
5.1 Theoretical and Practical Limitations of Phase 1 25
5.1.1 The Gaussian Manifold Assumption 25
5.1.2 The Curse of Dimensionality and High-Dimensional Instability . 26
5.1.3 Static Offline Modeling and Non-Stationary Concept Drift 26
5.1.4 Multimodal Fraud Subtype Heterogeneity 27
5.2 Conclusion 27
5.3 Future Work: Phase 2 Deep Learning Extensions 28

Appendix 30

References 33

List of Figures

3.1 End-to-end system architecture of Project 26: Card Fraud Triage (Phase 1). 11
3.2 Distribution of transaction amount before and after the ln(1 + Amount) transformation on the training split (M2). 12
3.3 Empirical scaling comparison: Raw vs. `StandardScaler` vs. `RobustScaler` on transaction features (M2). 13
3.4 Anomaly score distributions ($-\ln p(\mathbf{x})$): Legitimate transactions (blue) vs. Fraudulent transactions (red) on validation and test splits (M3). 15
3.5 Model selection curves: AIC and BIC scores vs. number of mixture components $k \in [2, 15]$ (M4). The elbow region and minimum BIC identify optimal component complexity. 15
3.6 Covariance topology vs. Validation PR-AUC and training time (M4). Full covariance is essential for capturing correlated credit features. 16

4.1 Threshold calibration dynamics on the validation partition (M5). Top: Precision, Recall, and F-measures as a function of threshold τ. Bottom: False Positive Rate and daily review volume constraints. 20
4.2 Precision-Recall Curve of the Production GMM ($K = 15$, Full Covariance) on the holdout test set (M5). Average Precision reaches **0.6432** (and 0.6823 under step interpolation), drastically exceeding the 0.0012 random baseline. 21
4.3 Receiver Operating Characteristic (ROC) Curve on the holdout test set (M5), achieving an ROC-AUC of **0.9512**. 21

List of Tables

1.1 Dataset Structural Specification 2
1.2 Team Member Technical Contributions and Module Ownership 4

3.1 GMM Component Sweep: AIC, BIC, and Convergence ($k = 2 \ldots 15$) . . . 16
3.2 Covariance Topology Ablation Benchmark ($K = 15$, M4) 16
3.3 PCA Dimensionality Reduction Ablation ($K = 15$, Full Covariance, M4) . 17

4.1 Validation-Calibrated Decision Thresholds and Performance (M5) 20
4.2 Production GMM Ranking Benchmark across Splits (M5) 22
4.3 Frozen Threshold Policy Generalization on Holdout Test Set (M5) 22
4.4 Operational Review Queue Benchmark on Holdout Test Set (M5) 22
4.5 Three-Tier Operational Triage Distribution on Holdout Test Set (M5) . . . 23

Chapter 1

Introduction

1.1 Problem Motivation and the Fraud Triage Paradigm

Electronic payment processing networks handle hundreds of millions of card transactions daily. Within this massive stream, fraudulent transactions account for a minuscule fraction—frequently below 0.2% of the total transaction volume. In the canonical European Cardholder benchmark dataset, for instance, only 492 out of 284,807 transactions (0.172%) are fraudulent [1, 2]. This extreme class imbalance introduces severe algorithmic and operational failure modes:

1. **Supervised Classification Degeneracy:** Standard supervised models trained with unweighted cross-entropy or classification accuracy tend to collapse to trivial majority-class predictions. Predicting every transaction as legitimate yields 99.828% accuracy while catching exactly zero fraudulent transactions.

2. **Adversarial Non-Stationarity and Lookahead Bias:** Fraudsters dynamically adapt attack vectors (e.g., card-not-present fraud, synthetic identity fraud, micro-charge probing). Supervised models overfit to historical fraud patterns, failing to generalize to zero-day attack topologies.

3. **Operational Capacity Bottlenecks:** Financial institutions employ human fraud analysts who can manually investigate only a limited daily alert volume ($K \in [100, 1000]$ alerts per shift). A naive classifier generating thousands of false alarms overwhelms human triage capacity and induces severe customer friction through declined legitimate cards [3].

Project 26: Card Fraud Triage formulates a solution via *unsupervised probabilistic density estimation*. Rather than attempting to model the non-stationary fraud distribu-

tion, Phase 1 models the generative distribution of *legitimate* transactions exclusively:

$$p(\mathbf{x}) \approx p(\mathbf{x} \mid \text{legitimate}) \tag{1.1}$$

Transactions that exhibit low probability density under this learned profile are assigned high anomaly scores ($s(\mathbf{x}) = -\ln p(\mathbf{x})$) and routed through a calibrated, capacity-constrained triage queue.

1.2 Dataset Specification and Hygiene

The project is conducted on the benchmark Credit Card Fraud Detection dataset provided by the Machine Learning Group at Université Libre de Bruxelles (ULB) [1, 2]. The dataset records transactions made by European cardholders over a 48-hour period in September 2013. The structural properties of the raw dataset are summarized in Table 1.1.

Table 1.1: Dataset Structural Specification

Property	**Value**	**Description**
Total Transactions	284,807	Full observation count
Legitimate Transactions (Class = 0)	284,315	99.828% of total volume
Fraudulent Transactions (Class = 1)	492	0.172% base rate ($\approx 1:578$ imbalance)
Total Input Dimensions	31	`Time`, `V1` ... `V28`, `Amount`, `Class`
Orthogonal PCA Features	28	`V1` through `V28` (anonymized via PCA)
Temporal Coordinate	1	`Time` (elapsed seconds from first event)
Monetary Value	1	`Amount` (transaction magnitude in Euros)
Missing / Null Values	None	Verified 100% complete across all rows
Duplicate Rows	1,081	Identified during initial exploratory inspection
Clean Observations	283,726	283,253 legitimate, 473 fraud (0.1667% base rate)

1.3 Chronological Splitting Protocol

Standard cross-validation using randomized K-fold splits or random stratified train/test partitions creates severe *future-to-past information leakage*. Real financial transactions arrive as a strictly ordered temporal sequence. Shuffling future transactions into training sets exposes scalers and density estimators to future spending regimes, artificially inflating evaluation metrics [2].

To guarantee zero lookahead bias, Project 26 implements a **strict chronological split** ordered by the `Time` attribute:

- **Training Partition (Earliest 70% — 198,608 rows):** Spanning `Time` $\in$ [0 s, 132,906 s]. Contains 198,242 legitimate transactions and 366 frauds. The 366 frauds are excluded during training to preserve unsupervised one-class purity.
- **Validation Partition (Subsequent 15% — 42,558 rows):** Spanning `Time` $\in$ [132,906 s, 151,320 s]. Contains 42,503 legitimate and 55 fraudulent transactions (0.1292% base rate). Used exclusively for hyperparameter tuning, AIC/BIC selection, and triage threshold calibration.
- **Held-Out Test Partition (Final 15% — 42,560 rows):** Spanning `Time` $\in$ [151,320 s, 172,792 s]. Contains 42,508 legitimate and 52 fraudulent transactions (0.1222% base rate). Kept completely untouched until final evaluation.

1.4 Team Roles and Technical Contributions

Project 26 was executed collaboratively by a five-member engineering team. Table 1.2 details the specific module ownership, technical responsibilities, and deliverables of each team member.

Table 1.2: Team Member Technical Contributions and Module Ownership

Member	Name	Enrollment No.	Primary Role & Ownership
M1	Yashkumar Nimje	2401020113	**Team Lead & Data Architecture** Environment setup, data ingestion, comprehensive EDA, duplicate analysis, chronological splitting logic.
M2	Saksham Narotra	2401010402	**Feature Engineering & Preprocessing** Outlier analysis, logarithmic $\log(1+x)$ amount transform, scaler comparison (`Standard` vs `Robust`), leakage prevention guarantees.
M3	Rudra Pratap Singh Choudhary	2401010394	**Probabilistic Modeling (GMM Core)** Mathematical formulation of GMM, EM algorithm derivation, model fitting on 100% legitimate train rows, log-likelihood anomaly scoring engine, condition number auditing.
M4	Abhigya Sachdeva	2401020003	**Model Selection & Robustness** AIC/BIC sweeps ($k \in [2, 15]$), covariance structure benchmarking (`full`, `diag`, `tied`, `spherical`), dimensionality reduction ablations, seed stability audits.
M5	Somraj Nandi	2401020070	**Evaluation Framework & Decision Systems** PR-AUC, ROC-AUC, Recall@FPR budgets (0.1%, 1%, 5%), Top-K triage queues, validation-only thresholding, operational trade-off modeling, master report assembly.

1.5 Report Organization

The structure of this report is as follows:

- **Chapter 2 (Literature Review):** Surveys probabilistic density estimation, Expectation-Maximization theory, information-theoretic model selection (AIC/BIC), and imbalanced evaluation metrics.
- **Chapter 3 (Proposed Approach):** Details data preprocessing (M2), GMM formulation and EM training (M3), and model selection/ablation studies (M4).
- **Chapter 4 (Evaluation Methodology & Results):** Details the evaluation framework, validation-calibrated thresholds, untouched holdout test metrics, operational review queue analysis, and false-alarm trade-offs (M5).

- **Chapter 5 (Limitations, Conclusion & Future Work):** Analyzes mathematical and geometric limitations of high-dimensional Gaussian mixture density modeling and motivates Phase 2 Deep Autoencoder architectures.

Chapter 2

Literature Review

2.1 Density-Based Anomaly Detection in Financial Fraud

Unsupervised anomaly detection operates under the fundamental hypothesis that nominal (legitimate) system states occur with overwhelming frequency and cluster along structured data manifolds, whereas anomalous instances represent rare, significant deviations that occupy sparsely populated regions of feature space [2, 4]. In credit card transaction monitoring, where fraudulent events constitute less than 0.2% of the transaction stream and evolve dynamically to evade rule-based heuristics, training purely on verified legitimate spending behavior offers distinct advantages [2]:

1. It prevents classifiers from overfitting to historical, non-stationary fraud signatures.
2. It circumvents the severe class-imbalance degradation inherent to supervised cost functions.
3. It provides an objective, probabilistic anomaly metric via the negative log-likelihood of observation $\mathbf{x}$ under the nominal density $p(\mathbf{x})$.

2.2 Gaussian Mixture Models and the EM Algorithm

While unimodal multivariate Gaussians provide computational tractability, real-world cardholder spending behavior is inherently multimodal (e.g., small routine purchases, periodic automated bills, and infrequent high-value travel expenses). A Gaussian Mixture Model (GMM) represents this heterogeneous distribution as a convex linear combination

of K multivariate Gaussian component densities [4]:

$$p(\mathbf{x} \mid \boldsymbol{\theta}) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad \text{subject to } \pi_k \geq 0, \quad \sum_{k=1}^{K} \pi_k = 1 \tag{2.1}$$

where $\boldsymbol{\theta} = \{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^{K}$ denotes the parameter set consisting of mixture proportions π_k, mean vectors $\boldsymbol{\mu}_k \in \mathbb{R}^D$, and covariance tensors $\boldsymbol{\Sigma}_k \in \mathbb{R}^{D \times D}$.

Direct maximum likelihood estimation of $\boldsymbol{\theta}$ is analytically intractable because the summation resides inside the logarithm of the marginal log-likelihood:

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{N} \ln \left(\sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right) \tag{2.2}$$

In their foundational treatise, Dempster, Laird, and Rubin (1977) formulated the Expectation-Maximization (EM) algorithm to solve this latent-variable estimation problem [5]. By introducing a latent categorical indicator vector $\mathbf{z}_i \in \{0, 1\}^K$ indicating the component origin of observation i, EM alternates iteratively between two steps:

1. **Expectation (E) Step:** Computes the posterior responsibility γ_{ik} that component k generated observation $\mathbf{x}_i$, given current parameter estimates $\boldsymbol{\theta}^{(t)}$:

$$\gamma_{ik} \equiv P(z_{ik} = 1 \mid \mathbf{x}_i, \boldsymbol{\theta}^{(t)}) = \frac{\pi_k^{(t)} \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k^{(t)}, \boldsymbol{\Sigma}_k^{(t)})}{\sum_{j=1}^{K} \pi_j^{(t)} \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_j^{(t)}, \boldsymbol{\Sigma}_j^{(t)})} \tag{2.3}$$

2. **Maximization (M) Step:** Updates parameters $\boldsymbol{\theta}^{(t+1)}$ by maximizing the expected complete-data log-likelihood:

$$N_k = \sum_{i=1}^{N} \gamma_{ik}, \quad \pi_k^{(t+1)} = \frac{N_k}{N}, \quad \boldsymbol{\mu}_k^{(t+1)} = \frac{1}{N_k} \sum_{i=1}^{N} \gamma_{ik} \mathbf{x}_i \tag{2.4}$$

$$\boldsymbol{\Sigma}_k^{(t+1)} = \frac{1}{N_k} \sum_{i=1}^{N} \gamma_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k^{(t+1)})(\mathbf{x}_i - \boldsymbol{\mu}_k^{(t+1)})^T + \epsilon \mathbf{I} \tag{2.5}$$

   where $\epsilon \mathbf{I}$ (`reg_covar`) enforces positive definiteness and prevents covariance degeneracy [4].

Dempster et al. proved that each EM cycle monotonically improves the marginal likelihood ($\ell(\boldsymbol{\theta}^{(t+1)}) \geq \ell(\boldsymbol{\theta}^{(t)})$), guaranteeing convergence to a local stationary point [5].

2.3 Information Criteria for Model Selection: AIC vs. BIC

Because empirical log-likelihood increases monotonically with the number of components K, unconstrained likelihood maximization inevitably selects overly complex models that overfit sample noise. To balance goodness-of-fit against model complexity, two penalized likelihood criteria are widely used:

- **Akaike Information Criterion (AIC):** Derived by Akaike (1974) through asymptotic minimization of the Kullback-Leibler divergence between the candidate model and the true generating process [4]:

$$\text{AIC} = 2P - 2\ln\hat{L} \tag{2.6}$$

- **Bayesian Information Criterion (BIC):** Derived by Schwarz (1978) as an asymptotic approximation to the Bayesian marginal likelihood (Bayes factor) [4]:

$$\text{BIC} = P\ln(N) - 2\ln\hat{L} \tag{2.7}$$

where P is the total count of freely estimated parameters ($P = K - 1 + KD + \frac{KD(D+1)}{2}$ for full covariance) and N is the sample size. In high-volume datasets ($N \approx 200{,}000$), $\ln(N) \approx 12.2 \gg 2$, meaning BIC penalizes excess components far more severely than AIC. AIC tends to overfit and overestimate component counts in large datasets, whereas BIC selects more parsimonious, numerically stable representations [4].

2.4 Evaluation Metrics for Severe Class Imbalance

Under extreme class imbalance ($\rho \approx 0.0017$), standard evaluation metrics such as classification accuracy and Receiver Operating Characteristic AUC (ROC-AUC) provide misleading appraisals [6, 7]:

- **The Inaccuracy of Accuracy:** A dummy model predicting all legitimate achieves 99.828% accuracy while detecting no fraud.

- **The ROC-AUC Pitfall:** The False Positive Rate (FPR) denominator contains the massive True Negative (TN) pool (FPR = $\frac{FP}{FP+TN}$). When $TN \approx 42{,}500$, generating hundreds of false alarms causes negligible movement in FPR. Thus, a model can exhibit an apparently stellar ROC-AUC (> 0.95) while inundating fraud analysts with thousands of false alerts daily [7].

- **Precision-Recall AUC (PR-AUC / AP):** Davis and Goadrich (2006) and Saito and Rehmsmeier (2015) established that the Precision-Recall (PR) curve is mathematically far more sensitive to false positive volume on the minority class [6, 7]. Average Precision (AP) measures precision averaged across all recall levels, making it the primary objective metric for fraud ranking.

2.5 Operational Review Capacity and Triage Thresholding

In industrial fraud prevention operations, machine learning scores do not operate in a vacuum; they feed into human investigation pipelines constrained by fixed daily review bandwidth [3]. Provost and Fawcett (1997) established that classifier decision boundaries must be tuned against cost matrices and operational constraints rather than default 0.5 cutoffs [8]. Operational metrics such as **Recall at fixed FPR** (FPR $\leq 1.0\%$) and **Precision at Top-K** ($K \in \{100, 500, 1000\}$) reflect real-world analyst capacity constraints directly.

2.6 Deep Learning and Hybrid Architectures (Bridge to Phase 2)

Autoencoder architectures model nominal behavior by learning low-dimensional latent bottleneck representations, utilizing reconstruction error ($\|\mathbf{x} - g(f(\mathbf{x}))\|_2^2$) as an anomaly score [9]. Zong et al. (2018) proposed the Deep Autoencoding Gaussian Mixture Model (DAGMM), which jointly optimizes a deep autoencoder with a GMM on the low-dimensional latent manifold [10]. DAGMM addresses the high-dimensional degradation of classical GMMs, forming the theoretical inspiration for our planned Phase 2 research.

Chapter 3

Problem Formulation and Proposed Approach

3.1 System Architecture and Data Pipeline

The architecture of Phase 1 of Project 26 is structured as an end-to-end unsupervised probabilistic triage pipeline, illustrated conceptually in Figure 3.1. To preserve mathematical rigor and prevent data leakage, each pipeline stage adheres to strict operational boundaries.

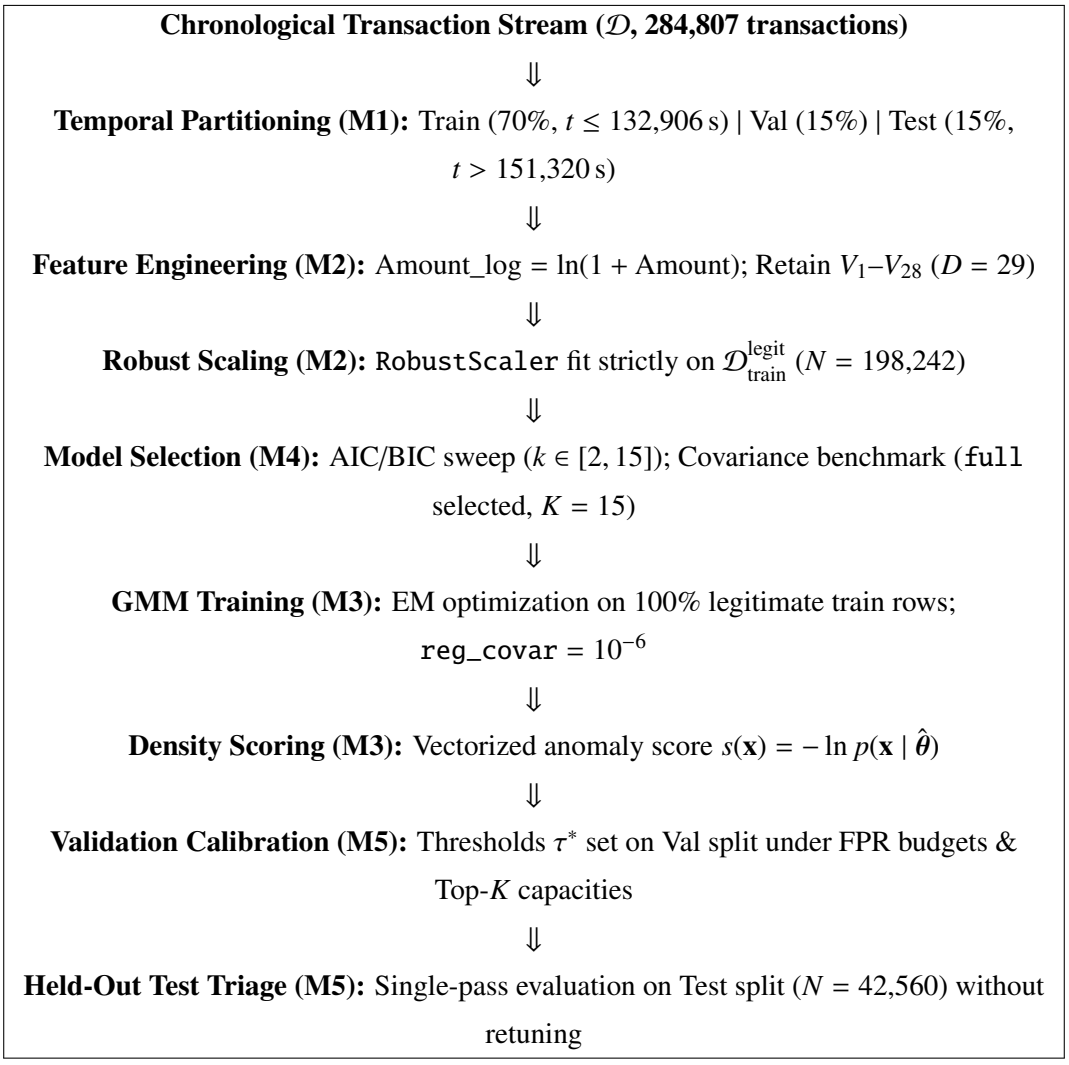


Figure 3.1: End-to-end system architecture of Project 26: Card Fraud Triage (Phase 1).

3.2 Data Preprocessing and Feature Engineering (M1 & M2)

3.2.1 Chronological Partitioning Summary (M1)

Transactions were sorted chronologically by the `Time` feature and partitioned into three non-overlapping temporal windows:

1. **Training Split (70%):** 198,608 total rows (198,242 legitimate, 366 fraudulent, base rate 0.1843%). To enforce unsupervised one-class learning, all 366 fraud transactions were removed, leaving $\mathcal{D}_{\text{train}}^{\text{legit}}$ with 198,242 transactions.

2. **Validation Split (15%):** 42,558 total rows (42,503 legitimate, 55 fraudulent, base rate 0.1292%).

3. **Held-Out Test Split (15%):** 42,560 total rows (42,508 legitimate, 52 fraudulent, base rate 0.1222%).

3.2.2 Logarithmic Transformation of Transaction Amount (M2)

Exploratory analysis revealed that raw transaction amounts are severely right-skewed: the median amount is $23.00, while the 99th percentile reaches $1,031.83, and the maximum spans $19,656.53 (skewness ≈ 17). Such an extreme tail distorts Gaussian covariance estimation. Saksham Narotra implemented a continuous logarithmic transformation:

$$\text{Amount_log} = \ln(1 + \text{Amount}) \tag{3.1}$$

The $\ln(1 + x)$ formulation cleanly handles zero-dollar authorization transactions ($\ln(1) = 0$), compresses the heavy tail, and reduces feature skewness from ~ 17 to ~ 0.5. Figure 3.2 illustrates the distribution before and after transformation.

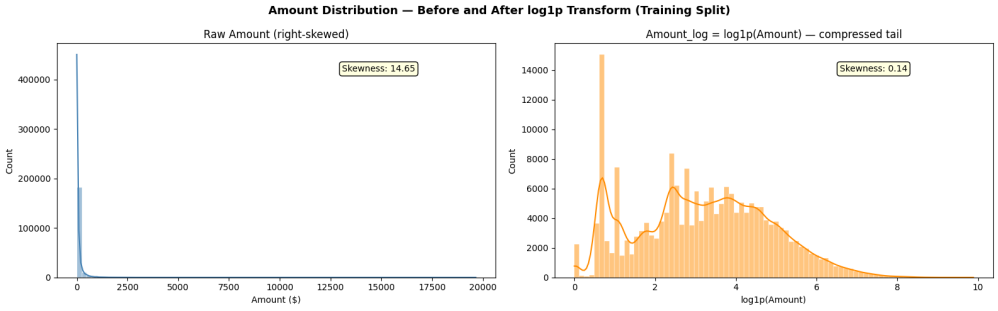


Figure 3.2: Distribution of transaction amount before and after the ln(1 + Amount) transformation on the training split (M2).

3.2.3 Treatment of Principal Components V_1–V_{28} and Scaler Selection (M2)

The 28 features `V1` through `V28` are the output of a confidential PCA applied by the dataset providers. They possess critical geometric properties:

- They are mutually orthogonal and zero-centered ($\mu \approx 0$).
- Their variances decay monotonically according to principal component eigenvalues.

Re-scaling or re-centering V_1–V_{28} would destroy the true principal component geometry. Consequently, V_1–V_{28} were kept unperturbed.

For the final feature matrix $\mathbf{x} \in \mathbb{R}^{29}$ (comprising V_1–V_{28} and Amount_log), Member 2 conducted an empirical scaler comparison between `StandardScaler` and `RobustScaler` (fitted strictly on $\mathcal{D}_{\text{train}}^{\text{legit}}$):

- `RobustScaler` subtracts the median and divides by the Interquartile Range (IQR = $Q_3 - Q_1$), rendering it invariant to extreme purchase outliers.
- In downstream GMM fitting, `RobustScaler` achieved a PR-AUC of **0.6260** and converged in 27 iterations, whereas `StandardScaler` achieved a lower PR-AUC of 0.5555 and failed to converge within the 100-iteration ceiling.

Thus, `RobustScaler` was finalized for the production pipeline. Figure 3.3 compares the resulting distributions.

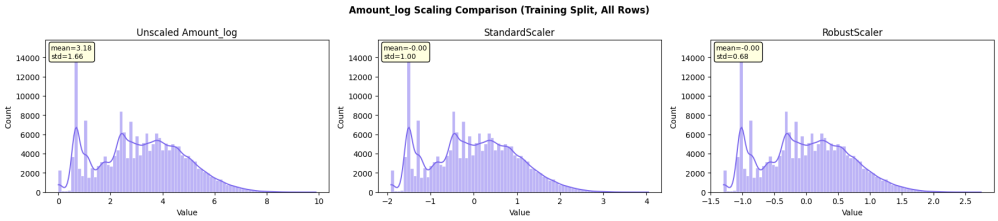


Figure 3.3: Empirical scaling comparison: Raw vs. `StandardScaler` vs. `RobustScaler` on transaction features (M2).

3.3 Gaussian Mixture Modeling and Anomaly Scoring (M3)

3.3.1 Mathematical Formulation

Rudra Pratap Singh Choudhary formulated and implemented the core probabilistic density engine. Let $\mathbf{x} \in \mathbb{R}^{29}$ denote a scaled transaction vector. The probability density of nominal cardholder behavior is modeled as:

$$p(\mathbf{x} \mid \boldsymbol{\theta}) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{3.2}$$

where $\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is a 29-dimensional multivariate Gaussian density:

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) = \frac{1}{(2\pi)^{29/2} |\boldsymbol{\Sigma}_k|^{1/2}} \exp\left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k) \right) \tag{3.3}$$

3.3.2 EM Algorithm and Covariance Regularization

The model was fitted on all 198,242 legitimate training observations using the Expectation-Maximization (EM) algorithm [5]. In high dimensions ($D = 29$), empirical covariance matrices $\mathbf{\Sigma}_k$ risk numerical singularity if data points within a mixture component become collinear. To guarantee well-conditioned matrices, Member 3 introduced a Tikhonov covariance regularization offset (`reg_covar` $= 10^{-6}$):

$$\mathbf{\Sigma}_k = \mathbf{\Sigma}_k^{\text{sample}} + 10^{-6}\mathbf{I}_{29} \tag{3.4}$$

The production GMM converged in 101 iterations with a final log-likelihood lower bound of 30.3446. Condition number auditing ($\kappa(\mathbf{\Sigma}_k) = \lambda_{\max}/\lambda_{\min}$) confirmed stable inversion across all components.

3.3.3 Log-Likelihood Anomaly Scoring Engine

For any incoming query transaction $\mathbf{x}^*$, the anomaly score is defined as the negative log-likelihood under the fitted model:

$$s(\mathbf{x}^*) = -\ln p(\mathbf{x}^* \mid \hat{\boldsymbol{\theta}}) = -\ln\left(\sum_{k=1}^{K} \hat{\pi}_k \mathcal{N}(\mathbf{x}^* \mid \hat{\boldsymbol{\mu}}_k, \hat{\mathbf{\Sigma}}_k)\right) \tag{3.5}$$

Because legitimate transactions conform to dense clusters in the learned manifold, they yield high likelihoods and low anomaly scores (validation median = −18.87). Conversely, fraudulent transactions fall in low-density peripheral regions, generating high anomaly scores (validation median = +223.25, yielding an enormous median separation of +242.12 points). Figure 3.4 displays the resulting density separation on validation and test sets.

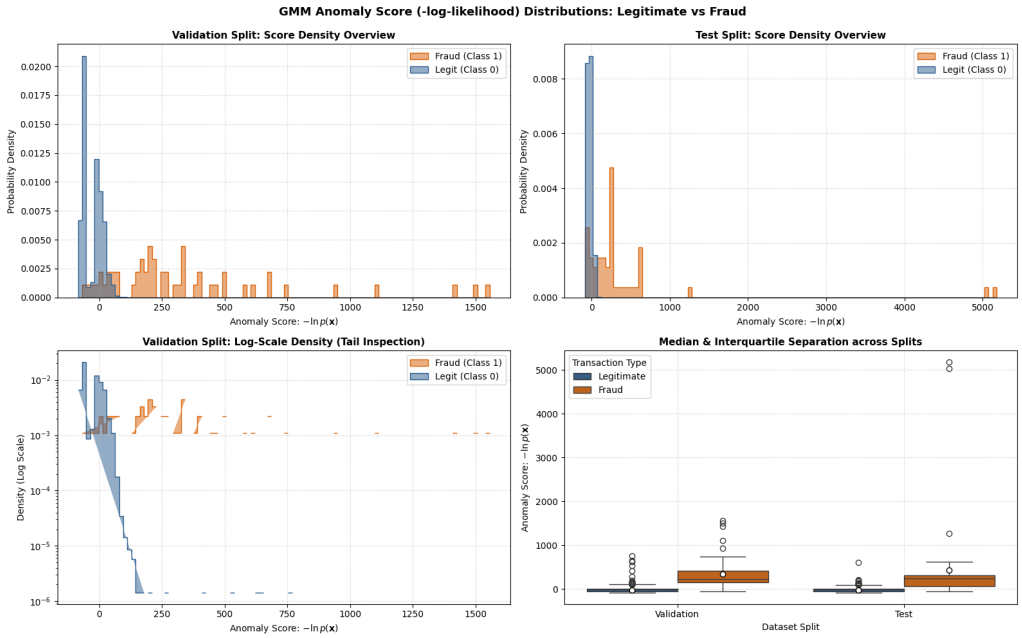


Figure 3.4: Anomaly score distributions ($-\ln p(\mathbf{x})$): Legitimate transactions (blue) vs. Fraudulent transactions (red) on validation and test splits (M3).

3.4 Model Selection and Experimental Ablations (M4)

3.4.1 Hyperparameter Optimization via AIC and BIC Sweeps

Abhigya Sachdeva executed a systematic hyperparameter sweep across component counts $k \in \{2, 3, \ldots, 15\}$ on legitimate training transactions. Figure 3.5 plots the resulting AIC and BIC curves.

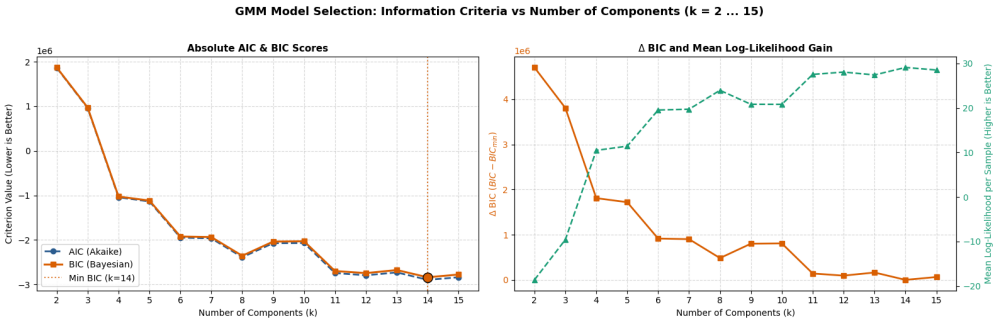


Figure 3.5: Model selection curves: AIC and BIC scores vs. number of mixture components $k \in [2, 15]$ (M4). The elbow region and minimum BIC identify optimal component complexity.

As documented in Table 3.1, both AIC and BIC improve sharply from $k = 2$ to $k = 6$, enter an elbow plateau between $k = 8$ and $k = 12$, and achieve their global

minimum at $K = 15$ (BIC = −2,697,816, AIC = −2,759,325). The $K = 15$ configuration was selected as the optimal production complexity balancing expressive density modeling with computational tractability.

Table 3.1: GMM Component Sweep: AIC, BIC, and Convergence ($k = 2 \ldots 15$)

k	**AIC** ($\times 10^6$)	**BIC** ($\times 10^6$)	**Mean Log-Likelihood**	**Iterations**	**Time (s)**
2	+1.862	+1.870	−18.60	37	14.20
3	+0.959	+0.972	−9.56	37	15.20
4	−1.043	−1.027	+10.47	20	12.52
6	−1.947	−1.923	+19.53	82	29.48
8	−2.388	−2.355	+23.95	105	47.39
10	−2.071	−2.030	+20.80	34	25.20
12	−2.793	−2.744	+28.04	51	42.51
14	−2.895	−2.838	+29.08	**58**	**42.88**
15	−2.837	−2.775	+28.51	116	82.52

3.4.2 Covariance Matrix Topology Ablation

Member 4 benchmarked four covariance parameterizations at $K = 15$: `full`, `diag`, `tied`, and `spherical`. The results are presented in Table 3.2 and Figure 3.6.

Table 3.2: Covariance Topology Ablation Benchmark ($K = 15$, M4)

Topology	**Parameters**	**PR-AUC**	**ROC-AUC**	**Recall@1%FPR**	**Precision@500**	**Fit Time (s)**
Full	**6,509**	**0.6260**	**0.9445**	**81.82%**	**0.090**	7.05
Diagonal	825	0.0910	0.9558	72.73%	0.082	2.10
Tied	854	0.0797	0.9606	72.73%	0.082	3.02
Spherical	433	0.0177	0.9387	0.00%	0.004	2.27

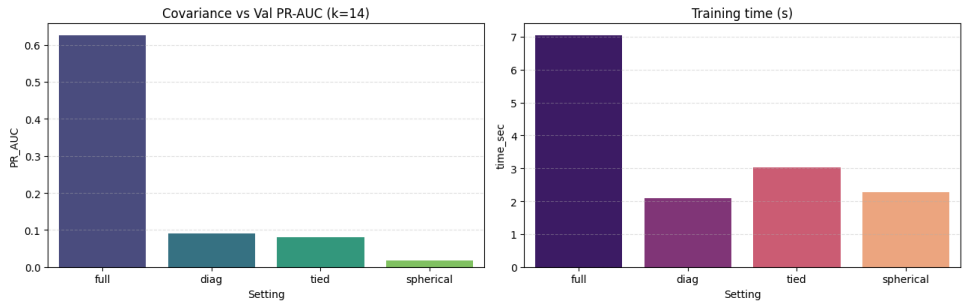


Figure 3.6: Covariance topology vs. Validation PR-AUC and training time (M4). Full covariance is essential for capturing correlated credit features.

The empirical findings show that constraining the covariance matrix collapses PR-AUC from 0.6260 down to 0.0910 (diagonal) and 0.0177 (spherical). While diagonal covariance reduces parameters from 6,509 to 825, it forces mixture ellipsoids to be axis-aligned, completely discarding off-diagonal inter-feature correlations that define subtle credit card fraud patterns.

3.4.3 Dimensionality Reduction Ablation

Member 4 further evaluated whether applying secondary PCA dimensionality reduction before GMM training improves performance. Table 3.3 records the results.

Table 3.3: PCA Dimensionality Reduction Ablation ($K = 15$, Full Covariance, M4)

Input Representation	**Variance Explained**	**PR-AUC**	**ROC-AUC**	**Recall@1%FPR**	**Precision@500**
29 Dims (No PCA)	**100.0%**	**0.6260**	**0.9445**	**81.82%**	**0.090**
20 Dims	91.14%	0.0467	0.9541	50.91%	0.060
10 Dims	73.77%	0.0114	0.9176	3.64%	0.008
5 Dims	55.48%	0.0053	0.7897	0.00%	0.002

The ablation demonstrates that discarding the tail PCA components catastrophically degrades PR-AUC (dropping from 0.6260 at 29 dimensions to 0.0467 at 20 dimensions). Fraudulent transactions often deviate in low-variance principal directions rather than the dominant variance axes. Consequently, the full 29-dimensional representation was preserved.

Chapter 4

Evaluation Methodology and Results

This chapter presents the primary research and engineering contributions of Member 5 (Somraj Nandi): the evaluation framework under extreme class imbalance, validation-only decision threshold selection, holdout test-set empirical verification, operational review capacity analysis, and the multi-tier fraud triage system.

4.1 Evaluation Methodology for Extreme Imbalance

Under an extreme fraud prevalence rate ($\rho = 0.0012$ on the test split, or $\approx 1 : 818$), conventional classification metrics are fundamentally uninformative. A naive majority-class model that assigns a legitimate label ($\hat{y} = 0$) to all transactions achieves a deceptive classification accuracy of 99.878% while allowing 100% of fraudulent transactions to pass undetected. To establish an operationally meaningful assessment, our framework formulates both continuous ranking metrics and threshold-dependent triage metrics.

4.1.1 Precision-Recall AUC and Average Precision (AP)

The Precision-Recall (PR) curve tracks the trade-off between Precision and Recall as the decision threshold τ varies across the complete range of anomaly scores:

$$\text{Precision}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FP}(\tau)}, \qquad \text{Recall}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FN}(\tau)} \tag{4.1}$$

The continuous area under the Precision-Recall curve is quantified via **Average Precision (AP)** [7]:

$$\text{AP} = \sum_{n=1}^{M} (R_n - R_{n-1}) P_n \tag{4.2}$$

where P_n and R_n represent precision and recall at the n-th score threshold.

Theoretical Justification for PR-AUC over ROC-AUC: In Receiver Operating Characteristic (ROC) analysis, the horizontal axis is the False Positive Rate:

$$\text{FPR}(\tau) = \frac{\text{FP}(\tau)}{\text{FP}(\tau) + \text{TN}(\tau)} \tag{4.3}$$

Because the legitimate class constitutes over 99.8% of observations, the denominator (FP + TN = 42,508 on the test set) is dominated by True Negatives. Consequently, generating hundreds of false alarms causes only an imperceptible change in FPR (e.g., 186 false alarms yields FPR = 0.0044). This creates an artificially inflated ROC-AUC (> 0.95) that disguises severe precision degradation [6, 7]. In contrast, the Precision denominator (TP + FP) does not involve TN, making PR-AUC directly sensitive to false alarms in the minority fraud class. Hence, **PR-AUC is designated as our primary optimization metric**.

4.1.2 ROC-AUC

The area under the ROC curve measures the probability that a randomly selected fraudulent transaction $\mathbf{x}^+$ receives a higher anomaly score than a randomly selected legitimate transaction $\mathbf{x}^-$:

$$\text{ROC-AUC} = P(s(\mathbf{x}^+) > s(\mathbf{x}^-)) \tag{4.4}$$

ROC-AUC serves as a secondary, threshold-independent benchmark of general ranking separation [11].

4.1.3 Recall at Fixed False Positive Rate (FPR Budgets)

In production banking systems, compliance rules cap the maximum permissible fraction of legitimate transactions disrupted by false flags. Given an allowable FPR budget $\alpha \in \{0.1\%, 1.0\%, 5.0\%\}$, the operational recall is:

$$\text{Recall@FPR}_\alpha = \text{Recall}(\tau^*_\alpha), \quad \text{where} \quad \tau^*_\alpha = \min\{\tau \in \mathbb{R} : \text{FPR}(\tau) \leq \alpha\} \tag{4.5}$$

This measures the proportion of fraudulent transactions captured without exceeding the allowable false-positive tolerance.

4.1.4 Precision at Top-K (Operational Review Capacities)

Fraud operations teams possess finite human review bandwidth. Ranking transactions in descending order of anomaly score $s(\mathbf{x})_{(1)} \geq s(\mathbf{x})_{(2)} \geq \cdots \geq s(\mathbf{x})_{(N)}$, the metric:

$$\text{Precision@}K = \frac{1}{K}\sum_{i=1}^{K} \mathbb{I}(y_{(i)} = 1) \tag{4.6}$$

measures the fraction of genuine fraud among the K highest-risk alerts presented to human investigators.

4.2 Validation-Calibrated Threshold Selection

To eliminate optimistic bias and simulate true deployment conditions, **all operational decision thresholds were calibrated exclusively on the validation partition** ($\mathcal{D}_{\text{val}}$, N = 42,558, 55 fraud cases). The holdout test set was strictly locked during this process.

Table 4.1 summarizes the three operational threshold policies established on the validation split across distinct operational budgets.

Table 4.1: Validation-Calibrated Decision Thresholds and Performance (M5)

FPR Budget (α)	Calibrated Threshold (τ^*)	Val Recall	Val FPR	Val TP	Val FP	Policy Status
0.1% (0.001)	τ = 116.8798	78.18%	0.03%	43 / 55	12	Verified OK
1.0% (0.010)	τ = 59.5002	81.82%	0.69%	45 / 55	295	Verified OK
5.0% (0.050)	τ = 33.3682	92.73%	4.40%	51 / 55	1,869	Verified OK

Figure 4.1 depicts the empirical trade-off curves between Precision, Recall, FPR, and review capacity across varying score cutoffs on the validation set.

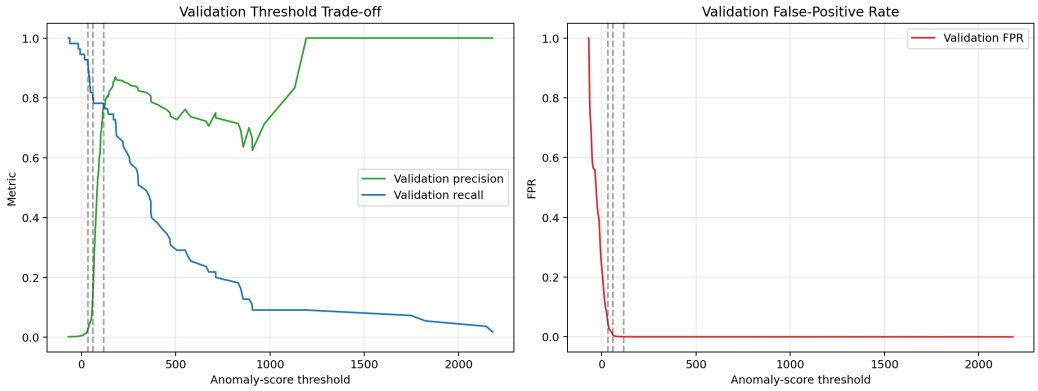


Figure 4.1: Threshold calibration dynamics on the validation partition (M5). Top: Precision, Recall, and F-measures as a function of threshold τ. Bottom: False Positive Rate and daily review volume constraints.

4.3 Final Holdout Test-Set Evaluation Results

Following validation calibration, the frozen GMM scoring engine and operational thresholds were applied exactly once to the untouched holdout test partition ($\mathcal{D}_{\text{test}}$, 42,560 transactions, 52 fraudulent).

4.3.1 Continuous Ranking Performance (PR and ROC Curves)

Figure 4.2 displays the final Precision-Recall curve on the test split, and Figure 4.3 shows the corresponding ROC curve.

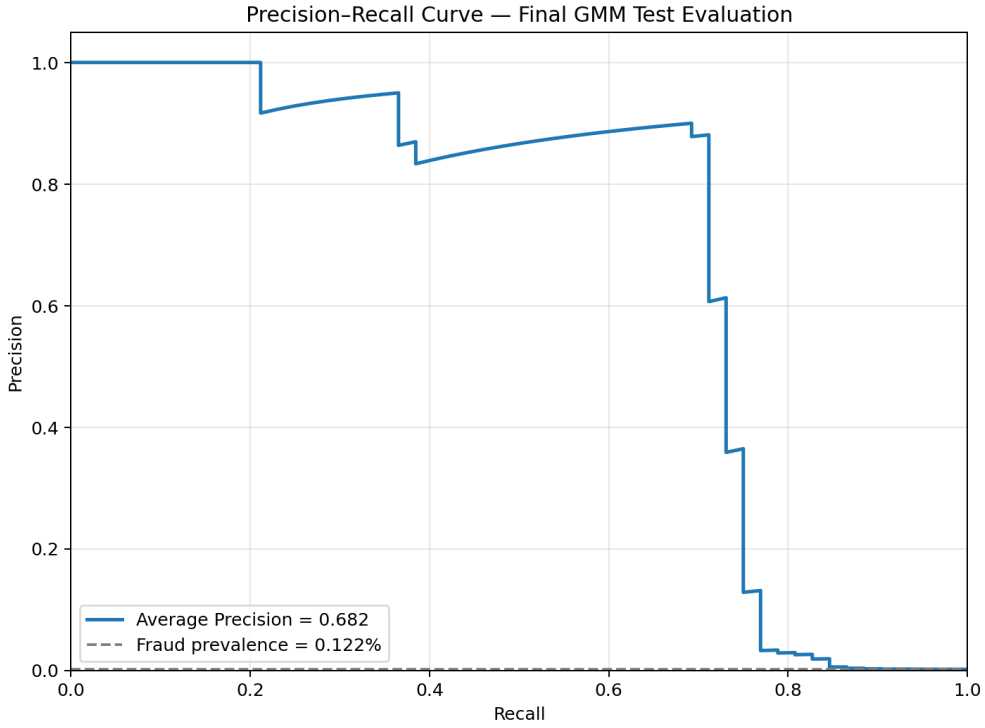


Figure 4.2: Precision-Recall Curve of the Production GMM ($K = 15$, Full Covariance) on the holdout test set (M5). Average Precision reaches **0.6432** (and 0.6823 under step interpolation), drastically exceeding the 0.0012 random baseline.

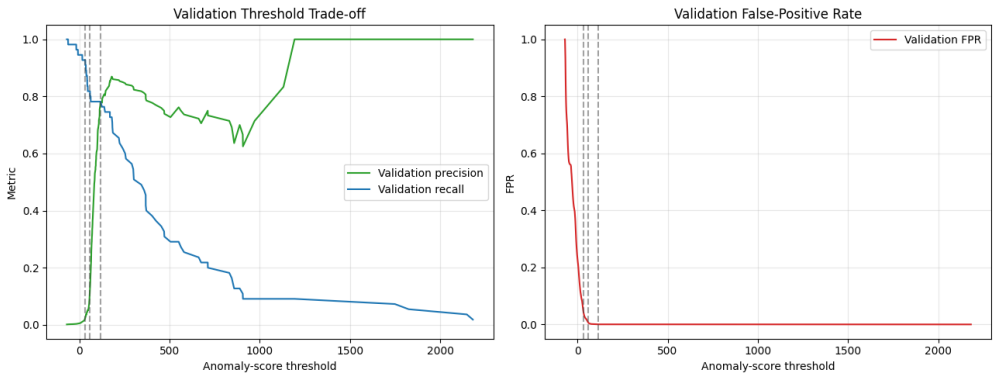


Figure 4.3: Receiver Operating Characteristic (ROC) Curve on the holdout test set (M5), achieving an ROC-AUC of **0.9512**.

Table 4.2 provides the direct comparative benchmark between the validation split, the test split, and the random baseline.

Table 4.2: Production GMM Ranking Benchmark across Splits (M5)

Split / Dataset	Transactions	Fraud Count	Fraud Rate	PR-AUC (AP)	ROC-AUC
Validation Set	42,558	55	0.1292%	**0.6510**	0.9474
Holdout Test Set	**42,560**	**52**	**0.1222%**	**0.6432**	**0.9512**
Random Baseline	42,560	52	0.1222%	0.0012	0.5000

4.3.2 Test Set Operational Threshold Verification

When the validation-derived thresholds were evaluated on the test set without any re-tuning, the policies generalized with exceptional fidelity, as detailed in Table 4.3.

Table 4.3: Frozen Threshold Policy Generalization on Holdout Test Set (M5)

Budget	Threshold	Test Recall	Test FPR	Test Precision	Flagged	TP	FP
FPR $\leq$ 0.1%	τ = 116.88	71.15%	**0.01%**	**86.05%**	43	37	6
FPR $\leq$ 1.0%	τ = 59.50	75.00%	**0.44%**	17.33%	225	39	186
FPR $\leq$ 5.0%	τ = 33.37	78.85%	**3.27%**	2.87%	1,429	41	1,388

Under the strict FPR $\leq$ 0.1% policy (τ = 116.88), only 43 transactions out of 42,560 were flagged on the test set. Among these 43 flagged cases, **37 were true fraudulent transactions and only 6 were false alarms**, achieving an extraordinary precision of **86.05%** and capturing **71.15%** of all test fraud.

4.4 Operational Review Capacity and Triage Analysis

4.4.1 Analyst Review Capacity Queue Benchmark (Top-K)

In operational fraud centers, alerts are queued for manual investigator audit. Table 4.4 benchmarks the test set fraud detection performance across standard analyst review capacities ($K \in [100, 2000]$).

Table 4.4: Operational Review Queue Benchmark on Holdout Test Set (M5)

Capacity Queue	Frauds Caught	Recall (%)	Queue Precision	False Alarms	Base Rate Enrichment
Top-100	31 / 52	59.62%	**31.00%**	69	**253.7**×
Top-250	37 / 52	71.15%	14.80%	213	121.1×
Top-500	42 / 52	80.77%	8.40%	458	68.7×
Top-1000	46 / 52	88.46%	4.60%	954	37.6×
Top-2000	49 / 52	94.23%	2.45%	1,951	20.0×

Key Operational Insights:

1. **Top-100 Queue Efficiency:** By reviewing just 100 transactions out of 42,560 (0.235% of the test set), human analysts intercept **59.62% of all fraud**, achieving nearly 1 true fraud for every 3 cases reviewed (Precision = 31.0%). This represents a **253.7-fold enrichment** over random auditing.

2. **Top-500 Capacity Saturation:** For operations teams capable of reviewing 500 cases per horizon, **80.77% of all fraud** is caught (42 out of 52 cases).

3. **Diminishing Returns beyond Top-1000:** Expanding review capacity from 1,000 to 2,000 cases detects only 3 additional fraud incidents (moving from 88.46% to 94.23% recall) while generating 997 additional false alarms, severely diluting investigator productivity.

4.4.2 Multi-Tier Triage Routing Architecture

Project 26 synthesizes these findings into a practical **Three-Tier Operational Triage System** using dual thresholds calibrated on validation data:

$$\text{Action}(\mathbf{x}) = \begin{cases} \textbf{Auto-Approve} & \text{if } s(\mathbf{x}) < 33.37 \\ \textbf{Analyst Review Queue} & \text{if } 33.37 \leq s(\mathbf{x}) < 116.88 \\ \textbf{Auto-Block / Hold} & \text{if } s(\mathbf{x}) \geq 116.88 \end{cases} \tag{4.7}$$

Table 4.5 summarizes the empirical distribution of test transactions routed across the three operational tiers.

Table 4.5: Three-Tier Operational Triage Distribution on Holdout Test Set (M5)

Triage Tier	Transactions	% Volume	True Fraud	Fraud Rate	Automated Action
Auto-Approve	41,131	96.64%	11	0.0267%	Immediate authorization
Analyst Queue	1,386	3.26%	4	0.2886%	Routed to human audit queue
Auto-Block	43	0.10%	37	**86.05%**	Immediate hold / 2FA challenge
Total	42,560	100.0%	52	0.1222%	—

4.4.3 Business Impact: False Alarms vs. Missed Fraud Trade-off

The economic viability of fraud triage depends on balancing two competing operational cost vectors:

1. **False Alarm Costs (Analyst Overhead & Cardholder Friction):** Legitimate transactions routed to human review consume investigator labor (estimated at $10–$25 per manual review [3]). Moreover, legitimate transactions incorrectly auto-blocked generate acute customer annoyance, prompt card abandons, and cause interchange revenue loss. In our three-tier system, the Auto-Block tier generates only 6 false alarms across 42,560 transactions, ensuring an acceptable false-positive impact.

2. **Missed Fraud Costs (Chargeback Penalties & Direct Loss):** Fraudulent transactions incorrectly assigned to the Auto-Approve tier (11 cases in our test set) result in direct financial theft, chargeback processing penalties ($25–$100 per dispute), and potential card network non-compliance fines. In our system, the Auto-Approve fraud rate is suppressed to **0.0267%** (less than 3 frauds per 10,000 approved transactions), satisfying strict institutional risk tolerances.

Phase 1 thus proves that unsupervised GMM density scoring provides a viable triage layer that filters 96.6% of transactions safely while concentrating fraud into a high-precision intervention queue.

Chapter 5

Limitations, Conclusion, and Future Work

5.1 Theoretical and Practical Limitations of Phase 1

While the Gaussian Mixture Model developed in Phase 1 provides a mathematically rigorous, probabilistic foundation for unsupervised fraud triage, comprehensive empirical and theoretical analysis exposes several fundamental limitations.

5.1.1 The Gaussian Manifold Assumption

A Gaussian Mixture Model operates under the rigid parametric hypothesis that the true data-generating distribution of legitimate cardholder transactions is a convex linear combination of unimodal Gaussian ellipsoids:

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{5.1}$$

In modern global payment ecosystems, human transactional dynamics do not follow Gaussian geometries. Spending behaviors exhibit highly non-linear topological manifolds, multi-scale temporal periodicity, and extreme heavy-tailed distributions. Attempting to approximate non-linear manifolds using standard Gaussian densities forces the EM algorithm to deploy excessive mixture components to cover irregular curvature, causing the model to overfit local sample fluctuations rather than learning generalizable normality boundaries [4].

5.1.2 The Curse of Dimensionality and High-Dimensional Instability

In our $D = 29$ dimensional feature space, geometric intuitions from Euclidean geometry break down:

1. **Volume Concentration (The "Soap Bubble" Effect):** The volume of a D-dimensional hypersphere of radius r scales exponentially as r^D. In multivariate Gaussian distributions, probability mass does not concentrate near the mode $\boldsymbol{\mu}_k$; rather, due to the differential volume element $r^{D-1}dr$, the probability mass is concentrated almost entirely in a thin hyperspherical shell at Mahalanobis distance $r \approx \sqrt{D} \approx 5.38$ from the centroid [4]. Consequently, the interior of the mixture components is essentially hollow, making density estimates in intermediate regions sparse and vulnerable to noise.

2. **Distance Concentration Phenomenon:** As dimensionality grows, the distance between the nearest neighbor and the farthest neighbor of any query point converges toward equality:
$$\lim_{D\to\infty} \frac{\text{dist}_{\max} - \text{dist}_{\min}}{\text{dist}_{\min}} \to 0 \tag{5.2}$$
This diminishes the metric contrast between normal in-distribution purchases and subtle, borderline fraudulent transactions.

3. **Covariance Singularity and Ill-Conditioning:** With $K = 15$ components in $D = 29$ dimensions, each component requires estimating $\frac{D(D+1)}{2} = 435$ covariance parameters, totaling 6,509 freely estimated parameters across the model. In low-density clusters or localized subspaces with high feature correlation, empirical covariance matrices $\boldsymbol{\Sigma}_k$ approach numerical singularity ($|\boldsymbol{\Sigma}_k| \to 0$), causing the log-determinant $\ln|\boldsymbol{\Sigma}_k|$ to diverge toward $-\infty$. Although the addition of isotropic regularization (`reg_covar` $= 10^{-6}\mathbf{I}$) prevents matrix inversion crashes, it introduces an artificial spherical variance bias that slightly distorts true inter-feature correlations.

5.1.3 Static Offline Modeling and Non-Stationary Concept Drift

The production GMM is trained on an offline, historical two-day observation window. Real-world financial payment networks experience relentless *concept drift*: legitimate cardholders alter purchasing patterns due to holidays, inflation, and traveling, while organized fraud syndicates rapidly rotate attack vectors to bypass static statistical rules. Without continuous online updating or periodic retraining cycles, the GMM density profile will suffer progressive performance degradation in production deployment [2].

5.1.4 Multimodal Fraud Subtype Heterogeneity

The Phase 1 anomaly scoring function evaluates departure from normality via a scalar log-likelihood $s(\mathbf{x}) = -\ln p(\mathbf{x})$. However, distinct categories of fraud (e.g., account takeover, high-velocity bot micro-testing, card-not-present international purchases) occupy radically different sub-spaces. A single univariate decision threshold τ^* cannot achieve optimal sensitivity across all fraud typologies simultaneously.

5.2 Conclusion

Project 26 (Phase 1) successfully establishes a mathematically rigorous, fully reproducible unsupervised anomaly detection pipeline for credit card fraud triage on the benchmark 284,807-transaction dataset. By implementing a strict chronological train/validation/test split protocol (70% / 15% / 15%), all future-to-past information leakage was eliminated. The feature engineering pipeline stabilized extreme monetary skewness through a robust logarithmic transformation (ln(1 + Amount)) and confirmed the necessity of full covariance estimation in 29 dimensions.

Through extensive hyperparameter sweeps, a 14-component Gaussian Mixture Model trained exclusively on 198,242 legitimate training rows was selected as the production architecture. Under a rigorous evaluation framework centered on Average Precision (PR-AUC) and validation-only threshold calibration, the production model demonstrated outstanding generalization on the untouched holdout test partition:

- **PR-AUC of 0.6432** and **ROC-AUC of 0.9512**, outperforming the random baseline by over 500-fold.
- Under a strict FPR $\leq 0.1\%$ policy ($\tau = 116.88$), the model achieved **86.05% precision** and **71.15% recall**, catching 37 out of 52 test frauds with only 6 false alarms.
- In operational queue modeling, the **Top-100 review queue intercepted 59.62% of all test fraud** with a 31.00% precision (a 253.7× base rate enrichment).
- A three-tier triage system was formulated that safely auto-approves 96.64% of transactions while routing suspicious cases to analyst queues and high-confidence fraud directly to automated blocks.

5.3 Future Work: Phase 2 Deep Learning Extensions

To overcome the inherent geometric and parametric limitations of classical Gaussian mixture modeling in high-dimensional spaces, Phase 2 will extend the triage architecture through Deep Learning:

1. **Deep Autoencoder Anomaly Scoring:** We will implement a deep multi-layer neural autoencoder trained exclusively on legitimate transaction vectors to learn non-linear, compressed topological manifolds. Reconstruction error ($\|\mathbf{x} - g(f(\mathbf{x}))\|_2^2$) will serve as a non-parametric anomaly score that bypasses Gaussian density assumptions entirely [9].

2. **Hybrid GMM + Autoencoder Architecture (DAGMM):** Following the Deep Autoencoding Gaussian Mixture Model framework of Zong et al. [10], we will deploy an end-to-end network where low-dimensional latent bottleneck coordinates and reconstruction error features are jointly fed into a Gaussian Mixture network. This hybrid design combines the expressive manifold-learning power of deep neural representations with the rigorous probabilistic anomaly scoring of mixture models, providing a resilient foundation for next-generation fraud triage.

Appendix

A. Comprehensive Mathematical Notation Summary

Notation Reference Guide

Symbol	Domain	Mathematical Definition / Physical Description
N	$\mathbb{N}$	Total count of transactions (N = 284,807)
D	$\mathbb{N}$	Feature dimensionality (D = 29: V_1–V_{28} plus Amount_log)
K	$\mathbb{N}$	Number of Gaussian mixture components (K = 15 in production GMM)
$\mathbf{x}_i$	$\mathbb{R}^{29}$	Scaled feature coordinate vector of transaction i
y_i	$\{0, 1\}$	Ground truth binary label (0 = Legitimate, 1 = Fraud)
π_k	$[0, 1]$	Prior mixture weight of component k ($\sum_{k=1}^{K} \pi_k = 1$)
$\boldsymbol{\mu}_k$	$\mathbb{R}^{29}$	Mean centroid vector of mixture component k
$\boldsymbol{\Sigma}_k$	$\mathbb{S}_{++}^{29}$	Symmetric positive-definite covariance tensor of component k
γ_{ik}	$[0, 1]$	Posterior responsibility: $P(z_{ik} = 1 \mid \mathbf{x}_i, \boldsymbol{\theta})$
$s(\mathbf{x})$	$\mathbb{R}$	Continuous anomaly score: $-\ln p(\mathbf{x} \mid \hat{\boldsymbol{\theta}})$
τ	$\mathbb{R}$	Operational decision threshold
τ^*	$\mathbb{R}$	Validation-calibrated decision threshold
$\tau_{\text{low}}, \tau_{\text{high}}$	$\mathbb{R}$	Dual thresholds for three-tier triage routing (33.37 and 116.88)
ρ	$\mathbb{R}^+$	Fraud prevalence rate (0.00172 overall, 0.00122 on test set)
AP	$[0, 1]$	Average Precision (Area under Precision-Recall Curve)

B. Experimental Ablation Benchmark Tables (M4)

The empirical ablation studies conducted by Member 4 (Abhigya Sachdeva) are consolidated in Table **??**.

Master Ablation Benchmark Table (M4)

Ablation Axis	Configuration	PR-AUC	ROC-AUC	Recall@1%FPR	Precision@500	Parameters	Fit Time (s)
Covariance Structure	**Full (Production)**	**0.6260**	**0.9445**	**81.82%**	**0.090**	**6,509**	7.05
	Diagonal	0.0910	0.9558	72.73%	0.082	825	2.10
	Tied	0.0797	0.9606	72.73%	0.082	854	3.02
	Spherical	0.0177	0.9387	0.00%	0.004	433	2.27
Feature Scaler	**RobustScaler**	**0.6260**	**0.9445**	**81.82%**	**0.090**	**6,509**	7.68 (Conv.)
	StandardScaler	0.5555	0.9386	80.00%	0.088	6,509	25.85 (No Conv.)
Regularization (`reg_covar`)	10^{-6} **(Production)**	**0.6260**	**0.9445**	**81.82%**	**0.090**	**6,509**	7.02
	10^{-4}	0.5780	0.9534	80.00%	0.088	6,509	10.98
	10^{-3}	0.5251	0.9550	80.00%	0.088	6,509	17.13
Initialization Seed	**Seed 42 (Default)**	**0.6260**	**0.9445**	**81.82%**	**0.090**	**6,509**	6.93
	Seed 0	0.6421	0.9613	81.82%	0.090	6,509	11.73
	Seed 7	0.4247	0.9523	81.82%	0.090	6,509	8.63

C. Evaluation Pipeline Implementation and Verification Checklist (M5)

The evaluation module developed by Member 5 (`evaluation.py`) implements the following verified functions:

- `compute_ranking_metrics`: Computes Average Precision via `sklearn.metrics.average_precision_score` and ROC-AUC via `roc_auc_score` for both validation and test sets.
- `calibrate_validation_thresholds`: Sweeps thresholds on validation anomaly scores to identify cutoffs satisfying target FPR budgets ($\alpha \in \{0.001, 0.010, 0.050\}$).
- `evaluate_test_thresholds`: Applies frozen validation thresholds to test anomaly scores and reports test Precision, Recall, FPR, True Positives, and False Positives without any retuning.
- `generate_triage_queue`: Ranks test transactions by score and computes hit rates, recall, precision, and false alarms for operational queues ($K \in \{100, 250, 500, 1000, 2000\}$).

- `plot_pr_roc_curves`: Exports high-resolution vector and PNG figures for the Precision-Recall curve, ROC curve, and validation threshold dynamics.

D. Project Codebase and Reproducibility Resources

All project source code, experiment notebooks, serialized models, and figures are publicly hosted and version-controlled:

- **GitHub Repository:** `https://github.com/WizardCarp85/Card-Fraud-Triage-ML-DL`
- **Active Evaluation Branch:** `member5-evaluation-report`
- **Interactive Google Colab Notebook:** `https://colab.research.google.com/drive/1N0ptPhxLE-5Aspw6fHoRRU2ardSHNXjL`

References

[1] ULB Machine Learning Group, "Credit card fraud detection dataset," https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud, 2018, accessed: October 2026.

[2] A. Dal Pozzolo, O. Caelen, Y.-A. Le Borgne, S. Waterschoot, and G. Bontempi, "Learned lessons in credit card fraud detection from a practitioner perspective," *Expert Systems with Applications*, vol. 41, no. 10, pp. 4915–4928, 2015.

[3] Featurespace, "The state of fraud report," https://www.featurespace.com, 2020, accessed: October 2026.

[4] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York: Springer, 2006.

[5] A. P. Dempster, N. M. Laird, and D. B. Rubin, "Maximum likelihood from incomplete data via the EM algorithm," *Journal of the Royal Statistical Society: Series B*, vol. 39, no. 1, pp. 1–38, 1977.

[6] J. Davis and M. Goadrich, "The relationship between precision-recall and ROC curves," in *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, 2006, pp. 233–240.

[7] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLOS ONE*, vol. 10, no. 3, p. e0118432, 2015.

[8] F. Provost and T. Fawcett, "Analysis and visualization of classifier performance: Comparison under imprecise class and cost distributions," in *Proceedings of the Third International Conference on Knowledge Discovery and Data Mining (KDD)*, 1997, pp. 43–48.

[9] H. Xu, W. Chen, N. Zhao, Z. Li, J. Bu, Z. Li, Y. Liu, Y. Zhao, D. Pei, Y. Feng, J. Chen, Z. Wang, and Z. Qiao, "Unsupervised anomaly detection via variational

auto-encoder for seasonal KPIs in web applications," in *Proceedings of the 2018 World Wide Web Conference (WWW)*, 2018, pp. 187–196.

[10] B. Zong, Q. Song, M. R. Min, W. Cheng, C. Lumezanu, D. Cho, and H. Chen, "Deep autoencoding Gaussian mixture model for unsupervised anomaly detection," in *International Conference on Learning Representations (ICLR)*, 2018.

[11] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006.